

Coherence Without Content

A value-coherence metric scores invented outcomes at 0.880

Nullcard

Apart Research Digital Minds Sprint

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The one-slide version

A published metric says these models have coherent values.

We ran it on outcomes that **mean nothing**.

real outcomes	0.906
meaningless outcomes	0.880
difference	+0.025

The metric is not broken. It is **unanchored**.

And the evidence was in the forward pass all along: the channel the metric **discards** tells real outcomes from nonsense at 0.821; the channel it **keeps** manages 0.596.

The claim we are testing

Utility Engineering (Mazeika et al., 2025)

- Ask a model to choose between two outcomes, 500 of them
- Fit a Thurstonian utility model to the choices
- Report held-out accuracy as *structural coherence*
- It rises with scale $\Rightarrow$ “value systems emerge”

Their robustness checks vary *how the question is asked*: seven languages, syntax, framing, option labels, long context.

No condition varies whether the outcomes mean anything.

Why that gap is reachable, not pedantic

High held-out accuracy proves choices follow a *stable scalar ordering*. It does not prove the ordering is *about* anything.

- **244 of 510 outcomes contain a numeral** — orderable by arithmetic alone. Whole categories are 100% numeric.
- In simulation, a responder ordering purely by magnitude, with **zero semantics**, scores **0.949–0.977** held out.
- A responder with a genuine latent utility function scores 0.858–0.928 — **lower**.

An instrument where “count the number” beats “have values” cannot tell them apart from its own output.

The control

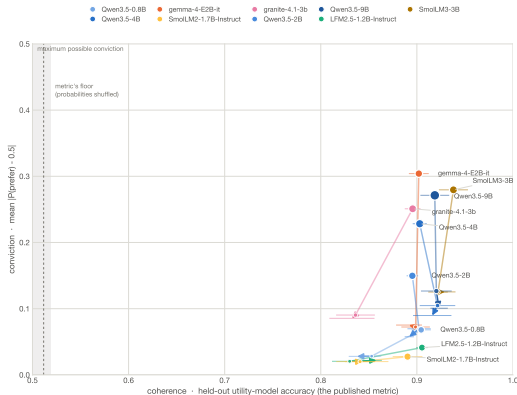
Same 510 outcomes. Same prompt, verbatim. Same fit. Same metric. **Only the referents change.**

arm	outcomes	coherence would reflect
R	real	values + arithmetic + format
N⁺	invented, magnitudes kept	arithmetic + format
N⁻	invented, magnitudes gone	format alone

N⁻ is the floor: what the procedure returns when there is nothing to have a preference *about*.

Every number we report is **R** – **N⁻**, never a raw coherence. 9 models, 81 cells, 3 design replicates each.

Result 1 — the metric barely notices



Each model is a *path*: real $\rightarrow$ invented-with-magnitudes $\rightarrow$ invented. If the metric tracked meaning, paths would run down-*and-left*. They run **almost straight down**.

Result 2 — direction survives, conviction does not

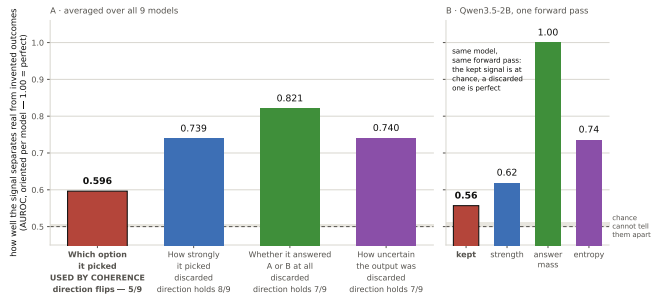
The metric thresholds preferences to hard labels. A pair at $p = 0.51$ counts exactly like a pair at $p = 0.99$.

commits to a side, real outcomes	41%
commits to a side, invented outcomes	4.5%
median collapse	17×

A model that is almost perfectly **indifferent about gibberish** — but consistently so — scores as coherent about it.

The metric discards the one channel that can see nonsense

Of four signals in one forward pass, coherence keeps the one that distinguishes real from invented outcomes least well



A taller bar means that signal separates the two arms more sharply — not that the model is better. AUROCs are oriented using known arm labels: they show how much separating information each channel carries, not the performance of a deployable detector.

Same forward pass, 9 models. Kept channel **0.596**; discarded answer mass channel **0.821**, up to 1.000 on Qwen3.5-2B.

The kept channel is binary, so a detection rate at a fixed 5% false-positive rate is not well defined for it; we compare AUROC only.

Result 3 — the scaling claim, floor-corrected

Pre-registered decisive test. One family, size the only variable, Qwen3.5-0.8B → Qwen3.5-9B.

- If values emerge with scale: floor stays flat, signal rises.
- If it is general answer-consistency: **the floor rises too**.

real-outcome coherence rises by	+0.014
the floor rises by	+0.085
floor-corrected residual	-0.071

Bigger models here are not more coherent *about the outcomes*. They are more consistent in general.

Pooled over all 9 models $r = -0.67$ ($p = 0.043$) — but that is *carried by this family* (within 4 sizes: -0.84; the other 5 models: -0.19). **We claim it within one family only.**

“Maybe your instrument just can’t see anything”

Fair objection. So: install a persona and measure the shift on **both arms**.

$$1 - \frac{\|\Delta_{\text{invented}}\|}{\|\Delta_{\text{real}}\|}$$

1.0 = pure preference change
0.0 = pure style change

- **18 of 20** conditions exceed +0.30, up to +0.85
- One informative exception at -0.66 — moved *gibberish* further than substance. That is what style looks like.
- **Depth barely matters**: user turn vs. system prompt shifts it 0.109; swapping the trait shifts it 0.276

So the instrument *can* see content. It is the unmanipulated coherence number that fails to depend on it.

Three checks that came out in their favour

We report these regardless. They **narrow** our claim.

1. **Order effects are handled.** Pure positional bias produces exactly 50/50 on counterbalanced pairs. It cancels completely.
2. **Not inflated by overfitting.** Held-out evaluation keeps a coin flip at 0.46–0.49. *Our first draft claimed otherwise and was wrong.*
3. **The fit is sound.** Shuffle probabilities across pairs and the metric lands at ~ 0.50 .

We are not saying the metric is broken.

We are saying **a high score does not distinguish values from format.**

What we would want you to hold against us

- **Tokenisation is not matched — so we tested it.** Invented outcomes carry $2.0\times$ the tokens. Matched on tokens (13–24), the residual is $+0.021$ vs $+0.025$ — it **survives**, and the length part runs against us.
- **Is the floor just length?** No — we checked. A one-parameter length rule gets 0.695 on the invented arm (vs 0.541 on real). But with length neutralised the fit still scores **0.848**. Models impose *rich* order on nonsense, not one cheap cue.
- **Scope.** 9 open-weight models, 0.8B–9B, one outcome set. The scaling claim rests on one family.
- **We shipped a bug and corrected it.** Six truncated result files reached an earlier analysis; one was 10% complete. Now every cell is row-count verified at two stages. Headline moved 0.002; one previously reported “instability” was the bug and is withdrawn.

A coherence number without a content control
does not distinguish **having values**
from **being consistent**.

The control costs one battery and a few dollars of GPU.
Reading the number without it costs more.

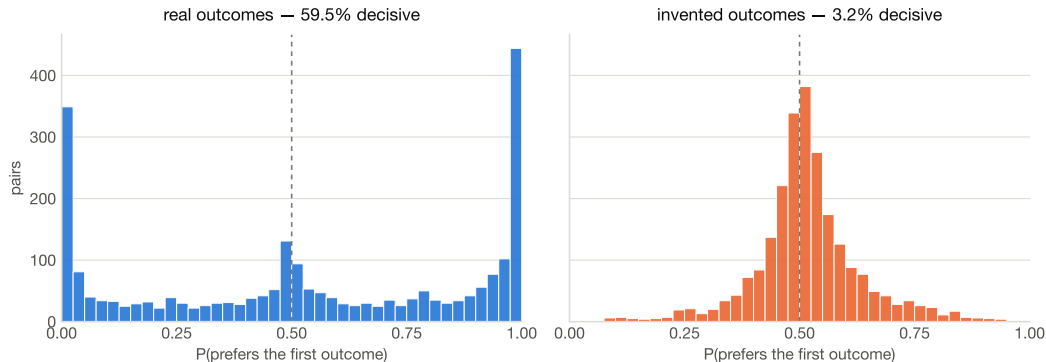
Pre-registered before the first run · battery hash-pinned 342db046213099ad... · code,
pre-registration and figures released.

Backup — all numbers

model	R	N ⁻	R-N ⁻	floor	clears	dec. R	dec. N ⁻
HuggingFaceTB/SmolLM3-3B	0.938	0.929	+0.009	0.029	no	48.7%	4.6%
Qwen/Qwen3.5-9B	0.919	0.923	-0.004	0.010	no	45.7%	2.7%
LiquidAI/LFM2.5-1.2B-Instruct	0.905	0.858	+0.047	0.023	2.0×	0.1%	0.0%
Qwen/Qwen3.5-0.8B	0.904	0.838	+0.067	0.023	2.9×	0.1%	0.0%
Qwen/Qwen3.5-4B	0.903	0.916	-0.013	0.015	no	33.8%	1.9%
google/gemma-4-E2B-it	0.902	0.892	+0.010	0.007	1.5×	56.4%	2.8%
ibm-granite/granite-4.1-3b	0.896	0.832	+0.063	0.029	2.2×	49.7%	14.9%
Qwen/Qwen3.5-2B	0.895	0.892	+0.003	0.001	> floor*	12.8%	0.0%
HuggingFaceTB/SmolLM2-1.7B-Instruct	0.890	0.843	+0.047	0.015	3.0×	0.0%	0.0%

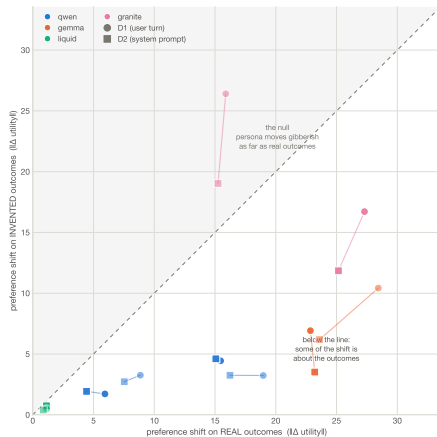
floor = range of the same cell across three independent designs. * = floor too near zero for a ratio to mean anything.

Backup — the mechanism, directly



Real outcomes: mass at the edges. Invented: mass at indifference. Coherence reads only *which side of 0.5*.

Backup — persona displacement



Dashed diagonal is the null. **Below** it = displacement the invented-outcome control cannot explain.